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JAVA INTERVIEW - CODING ROUND



TOP-25

TOP 25 DATABASE SQL FAQ

- 01 What is SQL?

- 02 What is a primary key?

- 03 What is a foreign key?

- 04 What are constraints in SQL?

- 05 Write a query to retrieve all records from a table named employees.

- 06 What is the difference between DELETE and TRUNCATE?

- 07 Write a query to fetch the maximum salary from employees.

- 08 Write a query to fetch the second-highest salary.

- 09 What is a JOIN? Explain its types.

- 10 Write a query to fetch employee names and department names using JOIN.

- 11 What is a GROUP BY clause?

- 12 Write a query to count employees in each department wise.



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TOP 25 DATABASE SQL FAQ

- 13 What is the difference between WHERE and HAVING clauses?

- 14 Write a query to fetch departments with more than 5 employees.

- 15 Explain UNION and UNION ALL.

- 16 What is a subquery in SQL?

- 17 Write a query to find all employees with a salary greater than the average salary.

- 18 Write a query to fetch the current date in SQL.

- 19 What is indexing in SQL?

- 20 What is normalization? Explain its types.

- 21 Write a query to add a new column to a table.

- 22 What is a stored procedure?

- 23 Write a stored procedure to fetch all employees.

- 24 Write a query to calculate the total salary for each department.

- 25 What are the ACID properties in SQL?



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1. What is SQL?

- SQL is a language used to manage and interact with databases.
- It helps to create, update, delete, and retrieve data.
- It is used with relational database systems like MySQL, Oracle, etc.

2. What is a primary key?

- A primary key uniquely identifies each row in a table.
- It ensures that no two rows have the same key value.
- A table can have only one primary key.

3. What is a foreign key?

- A foreign key connects two tables.
- It is a column in one table that refers to the primary key in another table.
- It ensures data consistency between related tables.

4. What are constraints in SQL?

- Constraints are rules applied to table columns.
- Examples: NOT NULL, UNIQUE, PRIMARY KEY, and FOREIGN KEY.
- They ensure data is valid and consistent.

5. Write a query to retrieve all records from a table named employees.

- Query:

```
SELECT * FROM employees;
```



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6. What is the difference between DELETE and TRUNCATE?

- DELETE removes selected rows based on a condition.
- TRUNCATE removes all rows and cannot be rolled back.
- DELETE activates triggers; TRUNCATE does not.

7. Write a query to fetch the maximum salary from employees.

- Query:

```
SELECT MAX(salary) FROM employees;
```

9. What is a JOIN? Explain its types.

- JOIN combines rows from two or more tables.
- Types:
 - INNER JOIN: Matches rows in both tables.
 - LEFT JOIN: All rows from the left table, matching rows from the right.
 - RIGHT JOIN: All rows from the right table, matching rows from the left.
 - FULL JOIN: Rows from both tables, even if no match.

10. Write a query to fetch employee names and department names using JOIN.

Query:

```
SELECT e.name, d.department_name  
FROM employees e  
JOIN departments d  
ON e.department_id = d.id;
```



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11. What is a GROUP BY clause?

- GROUP BY groups rows with the same value.
- It is used with aggregate functions like COUNT, SUM, or AVG.
- It helps summarize data.

12. Write a query to count employees in each department.

```
SELECT department_id, COUNT(*)  
FROM employees  
GROUP BY department_id;
```

13. What is the difference between WHERE and HAVING clauses?

WHERE filters rows before grouping.

- HAVING filters groups after grouping.
- HAVING is used with aggregate functions like COUNT or SUM.

14. Write a query to fetch departments with more than 5 employees.

```
SELECT department_id, COUNT(*)  
FROM employees  
GROUP BY department_id  
HAVING COUNT(*) > 5;
```



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15. **Explain UNION and UNION ALL.**

- UNION combines results of two queries, removing duplicates.
- UNION ALL includes duplicates in the results.
- Both are used to merge data.

16. **What is a subquery in SQL?**

- A subquery is a query inside another query.
- It helps retrieve data for the main query.
- It is commonly used in WHERE or SELECT clauses.

17. **Write a query to find all employees with a salary greater than the average salary.**

```
SELECT *  
FROM employees  
WHERE salary > (SELECT AVG(salary) FROM employees);
```

18. **Write a query to fetch the current date in SQL.**

- Query: `SELECT CURRENT_DATE;`



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19. What is indexing in SQL?

- Indexing makes searching for rows faster.
- It is a data structure created on table columns.
- Commonly used on primary and frequently searched keys.

20. What is normalization? Explain its types.

Normalization organizes data to reduce redundancy.

- Types:
 - 1NF: Removes duplicate columns.
 - 2NF: Eliminates partial dependencies.
 - 3NF: Removes transitive dependencies.

21. Write a query to add a new column to a table.

```
ALTER TABLE employees  
ADD COLUMN email VARCHAR(255);
```



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22. What is a stored procedure?

- A stored procedure is a reusable set of SQL commands.
- It is stored in the database and executed as needed.
- It improves efficiency and reusability.

23. Write a stored procedure to fetch all employees.

- Query:

```
CREATE PROCEDURE GetAllEmployees()  
BEGIN  
    SELECT * FROM employees;  
END;
```

24. Write a query to calculate the total salary for each department.

- Query:

```
SELECT department_id, SUM(salary)  
FROM employees  
GROUP BY department_id;
```

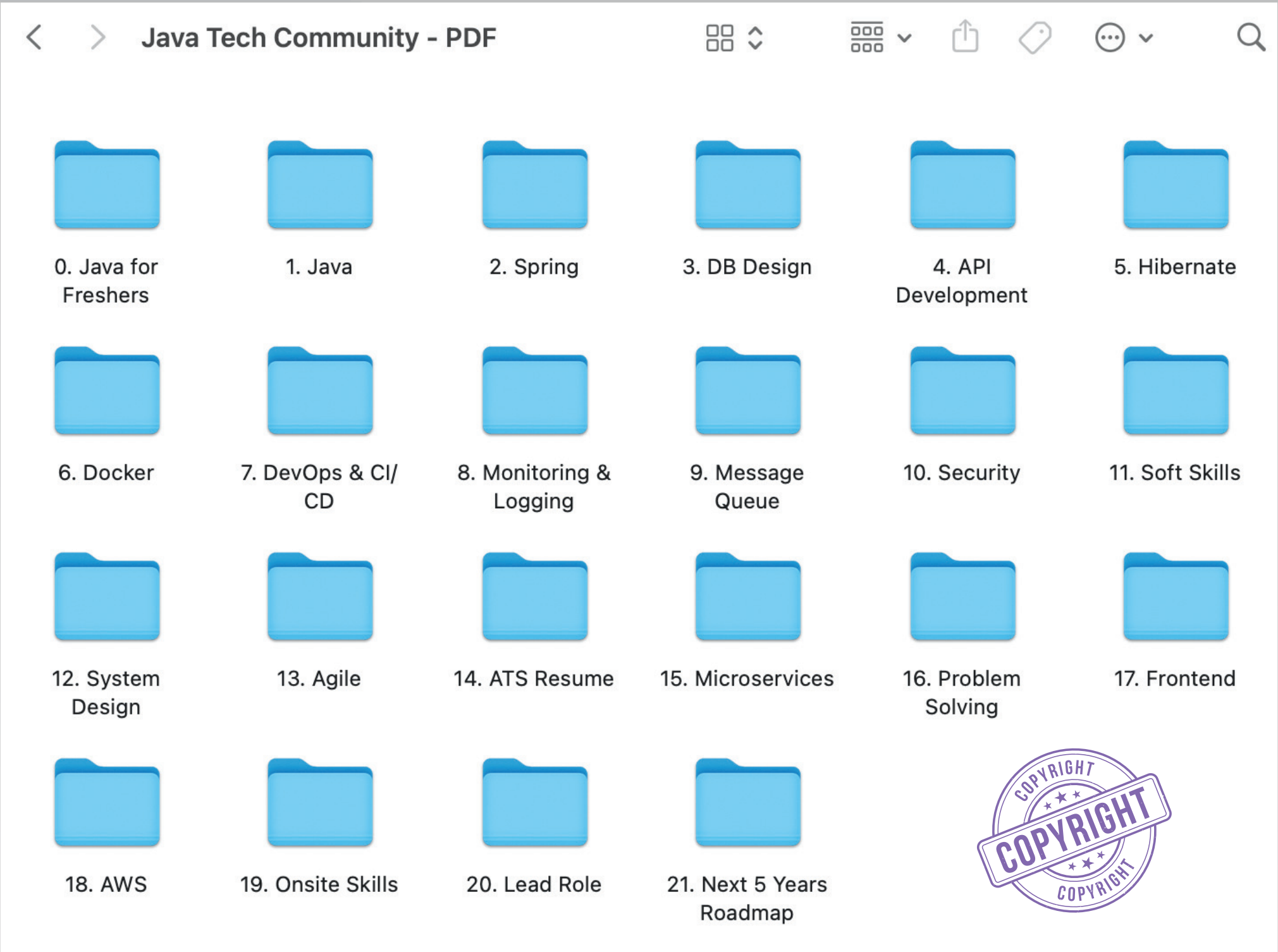
25. What are the ACID properties in SQL?

- ACID ensures reliable transactions.
 - Atomicity: All actions succeed or fail as a unit.
 - Consistency: Keeps data valid before and after a transaction.
 - Isolation: Transactions do not affect each other.
 - Durability: Changes are permanent after a transaction.

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